Jakub Dranczewski

I am a Research Assistant in Physics at Imperial College London,
with a passion for experimental investigation and threading interesting theory into it.

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Research Work Experience

2025- **Research Assistant**, *Imperial College London*
PI: Dr. Jack C. Gartside, Co-I: Prof. Riccardo Sapienza

- President's Excellence Fund for Frontier Research grant: "Hybrid Magneto-Photonic Neuromorphic Networks" (HYMN).
- Continuing PhD work on physical computing, expanded to tuning the network lasers and hybrid approaches utilising multiple systems.

2021-2025 **PhD student**, *IBM Research Europe - Zürich & Imperial College London*
Supervisors: Prof. Kirsten Moselund, Prof. Riccardo Sapienza, Heinz Schmid

- H2020 Marie Skłodowska-Curie Innovative Training Networks project (GA no. 859841): "COntrolling network RAndom Lasers on chip" (CORAL).
- First two years at IBM Research Europe - Zürich, developing plasma etching for InP and quantum well nanostructures, and nanolaser characterisation methods.
- Remaining time at Imperial College London, characterising samples prepared at IBM and developing physical machine learning experiments with on-chip network lasers.

Education

2017-2021 **MSci Physics student**, *Imperial College London*

- First-Class Honours, graduated top of the class, grade: 82.48%
- Masters thesis: *Time-Varying and Nonlinear Effects in an Indium Tin Oxide Nanolayer*
- Awarded the Abdus Salam Undergraduate Prize, Governors' MSci Prize in Physics, Ken Allen Prize for Academic Excellence (twice) and the Richard Learner Prize for Excellence in Second Year Laboratory.

2016-2017 **A Levels course**, *Dulwich College London*

- Physics, Mathematics, Further Mathematics, Computer Science; A*A*A*A*.
- Course accelerated, completed in a year as a scholarship organised by the Polish Children's Fund.

Other Research Experience

2.07 – 29.07.2020 **Remote Undergraduate Research Opportunities Programme placement**, *Imperial College London*
Supervisor: Prof. Roland A Smith

- Produced a ray tracing solution for predicting optical trapping of arbitrarily-shaped targets.
- Studied ray and wave optics, accelerating Python code, and quaternion-based rotational dynamics.

5.08 – 27.09.2019 **Undergraduate Research Opportunities Programme placement with the Experimental Solid State group**, *Imperial College London*
Supervisors: Prof. Riccardo Sapienza, Dr Stefano Vezzoli

- Gained understanding of multiple concepts in nonlinear optics and nanostructure design.
- Designing and preparing experimental equipment, procedures, and software, taking measurements, data analysis, and applying existing theoretical models; contributed to a publication.

6.08 – 28.09.2018 **Undergraduate Research Opportunities Programme placement with the Plasma Physics Group**, *Imperial College London*
Supervisor: Dr Jack Hare

- Developed *Magic2*, a GUI programme for interferometry data processing; contributed to a publication.
- Maintenance work on the Mega Ampere Generator for Plasma Implosion Experiments (MAGPIE).

2016, 2017 **Research Internships in the Institute of Physics of the Polish Academy of Sciences**, *organised through the Polish Children's Fund*
Supervisors: Dr Łukasz Kłopotowski, MSc Julia Miłosz, MSc Zygmunt Miłosz

- Three placements in two laboratories. Measuring photoluminescence decays and spectra of quantum dots; imaging graphite with a scanning tunneling microscope.

Skills

Programming — **Fluent in Python (numpy/scipy, matplotlib, data analysis, graphical interfaces – author of *puzzlepiece.readthedocs.io*)**, web development (JavaScript, PHP, MySQL), LaTeX, basic experience with C, C++, LabView, Matlab, and Origin Pro.

Electronics — **Experience working with the Arduino and ESP32 platforms, Raspberry Pi computers and microcontrollers**, low-level microprocessor programming, and electronic circuits.

Experiments — **Worked with optical table equipment, short and high-energy laser pulses, oscilloscopes and signal generators, time tagged time-resolved data collection, computer measurement systems, and advanced imaging equipment (SEM, STM).**

Nanofabrication — **Experience working in a research cleanroom environment**, work with plasma etching, material deposition, e-beam and optical lithography.

Teaching — Laboratory demonstrating, university programming helpdesk, private tutoring of A Level students.

Languages — **English**, advanced (IELTS mark 8.5/9, 2016); **Polish**, native speaker; **German**, basic.

Achievements and Awards

2020 — **International Research Opportunities Programme placement at the Massachusetts Institute of Technology (MIT)**, *including bursary of £5600*, cancelled due to Covid-19, part of funds granted for a remote UROP project at Imperial College

2019 — **EPSRC Vacation Bursary**, *£2581*, funding for the UROP project with the Experimental Solid State Group

2018, 2019 — **Finalist (2018) and Runner-up (2019) in the Royal College of Science Union *Science Challenge***, for popular science videos on quantum algorithmics and strong AI

2016–2017 — **During the year in Dulwich College:**

- Finalist of the *BAFTA Young Game Designers Game Making Award*
- Gold and a Top 50 mark in the second stage of the *British Physics Olympiad*
- Team captain of team UK in the *International Young Physicists' Tournament* 2017 in Singapore
- Finalist of the *UK Bebras Computational Thinking Challenge*

2014–2016 — **Scholarship of the Marshal for the Lubusz Voivodeship (twice), The *Pasjopolis* scholarship**, *total of £1300*, awarded to students showing the best academic performance, future prospects, and passion for their field

Selected Publications

See all on Google Scholar

2025 — **Few-Shot Retinomorphic Vision in a Nonlinear Photonic Network Laser**, *J Dranczewski*, WK Ng*, A Fischer*, TV Raziman et al.*, preprint in review, arXiv:2407.15558

2024 — **Designed Semiconductor Network Random Lasers**, *D Saxena, A Fischer, J Dranczewski, WK Ng et al.*, Laser & Photonics Rev., 2024, 19, 2400623

2023 — **Plasma etching for fabrication of complex nanophotonic lasers from bonded InP semiconductor layers**, *J Dranczewski, A Fischer, P. Tiwari, M. Scherrer et al.*, Micro and Nano Engineering, 2023, 19, 100196

2022 — **Saturable Time-Varying Mirror Based on an Epsilon-Near-Zero Material**, *R Tirole, E Galiffi, J Dranczewski, T. Attavar et al.*, Phys. Rev. Applied, 2022, 18, 054067

2019 — **Two-colour interferometry and Thomson scattering measurements of a plasma gun**, *J D Hare, J MacDonald, S N Bland, J Dranczewski et al.*, Plasma Phys. Control. Fusion, 61, 085012

Volunteering and Interests

2019-2021 — **Committee Member of the Imperial College Dramatic Society**

- Production manager and lighting designer for multiple shows and events. Skills in detailed planning, team management, budgeting, creativity, and ability to work with a large variety of equipment.

2020-2021 — **Imperial College Science Fiction and Fantasy Society Systems Administrator**

2014-2017 — **Member of the *Młodzi Lokalni (Young Locals)* voluntary association**

- Web development, graphic design, part of organising teams for city-wide events.

Hobbies — New technologies, photography, science-fiction and fantasy, art and poetry, cycling.